Asymmetric Synthesis of Jacobsen's Ligand

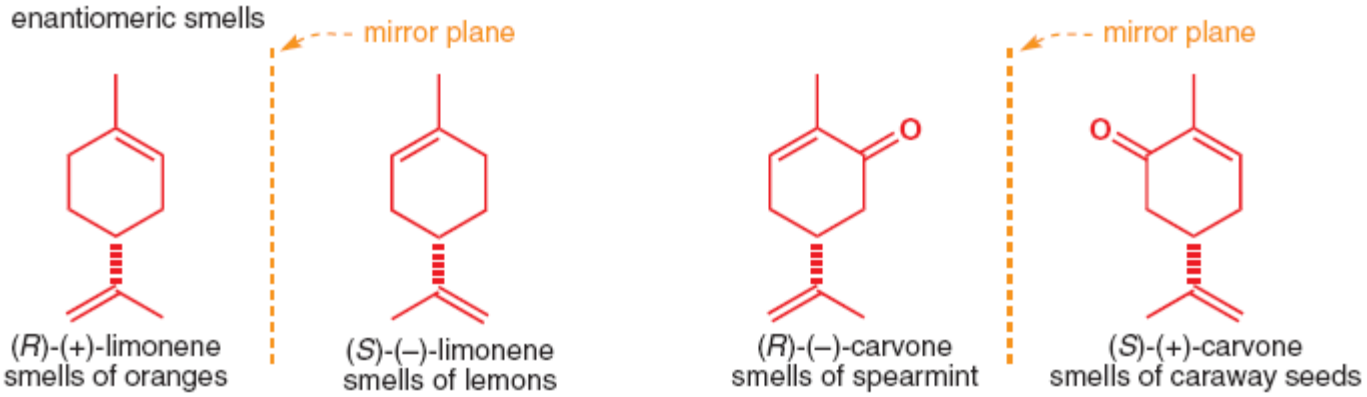
Dr. Zhang Sheng

Part 1

Theoretical Background

Chirality

- Chirality in nature

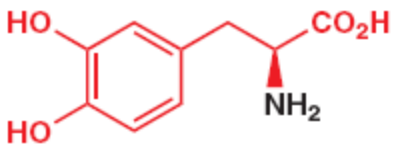


- All living systems are chiral
  - Amino acids
  - Sugars
  - DNA & RNA

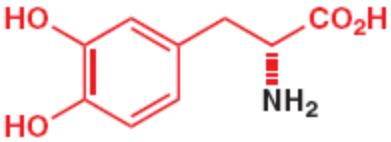
↳ can't use column, distillation, filtration, recrystalisation as enantiomers have same physical property

Chirality

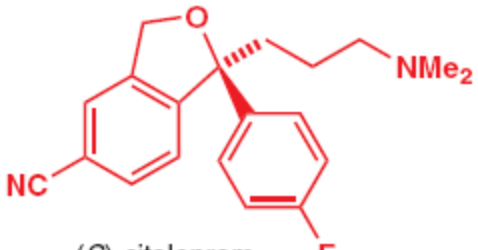
- Chiral drugs



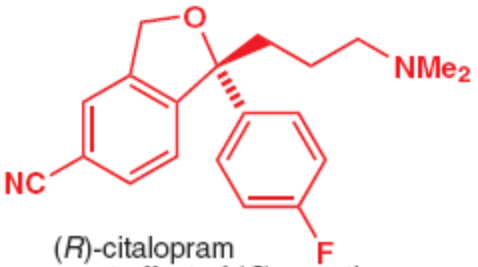
L-dopa
treatment for
Parkinson's disease



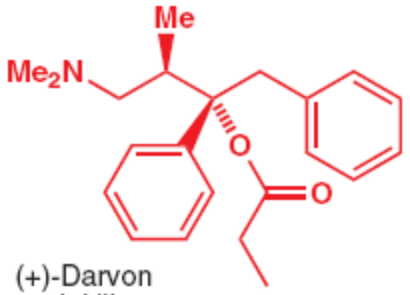
D-dopa
is toxic



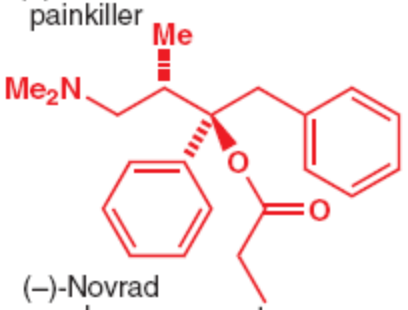
(*S*)-citalopram
drug for depression



(*R*)-citalopram
<1% antidepressant effect of (*S*) enantiomer

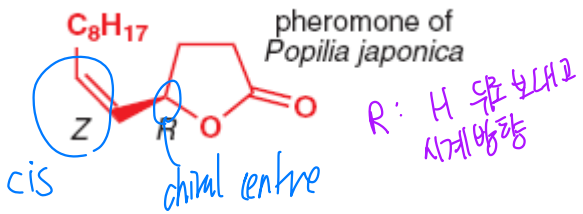


(+)-Darvon
painkiller



(–)-Novrad
cough suppressant

- Chiral pesticides

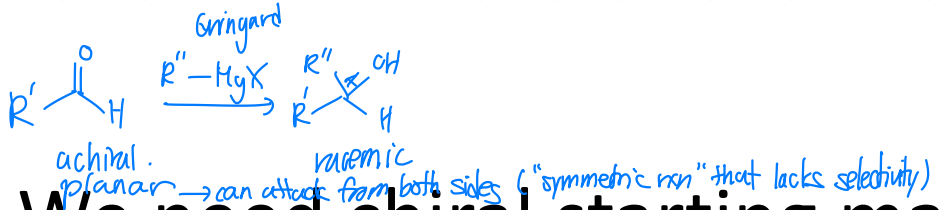


pheromone of
Popilia japonica

- For control of Japanese beetle:
  - 25 µg of (R) can attract thousands of beetles;
  - (R) with only 1% of (S) inhibits the activity i.e. cannot work as pesticide.
    ⇒ need high isomer purity

Where does chirality come from?

- Synthesis of a chiral compound from achiral or racemic starting materials always gives a racemic mixture of enantiomers.



- We need chiral starting materials, chiral reagents or chiral catalysts.

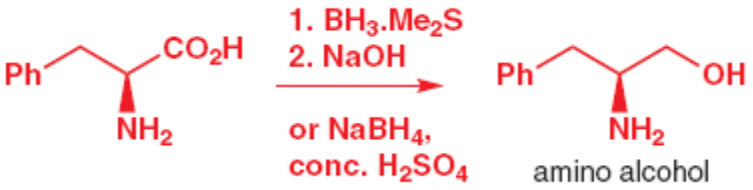
most effective method.
reagents need stoichiometric amount
while catalyst only needed by ~1 mol%.

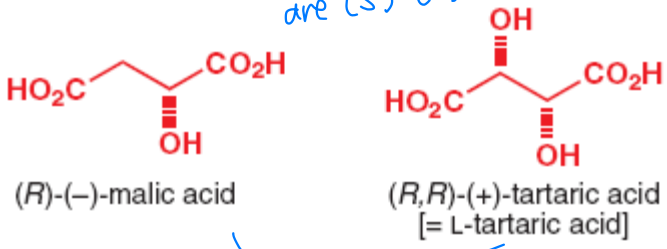
How to obtain these chiral molecules?

- Great gift from Nature
- Chiral pool (major categories)
  - Amino acids and derivatives



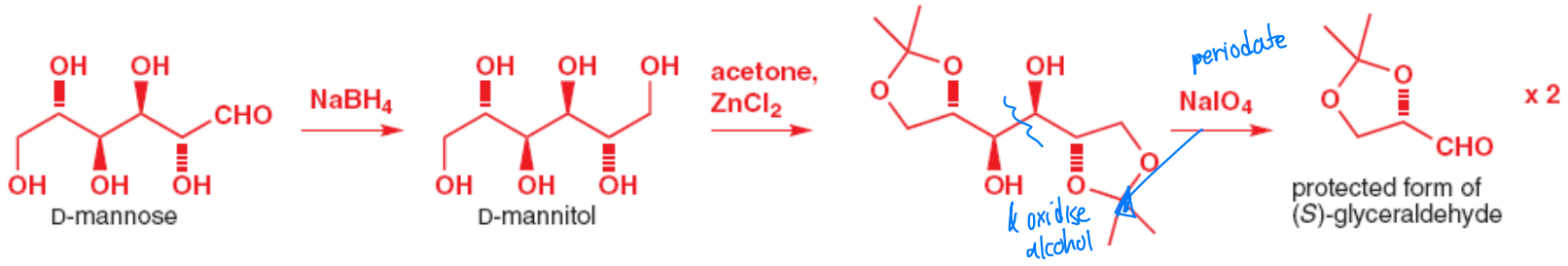
all natural aa are (S)-(–)





hydroxyl acid

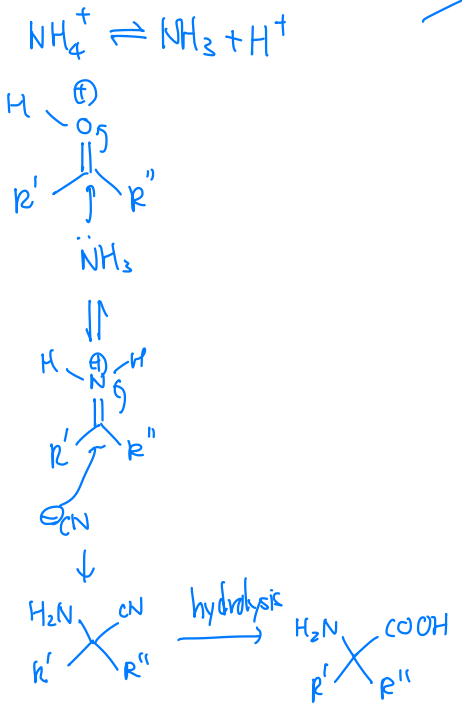
  - Carbohydrates and derivatives

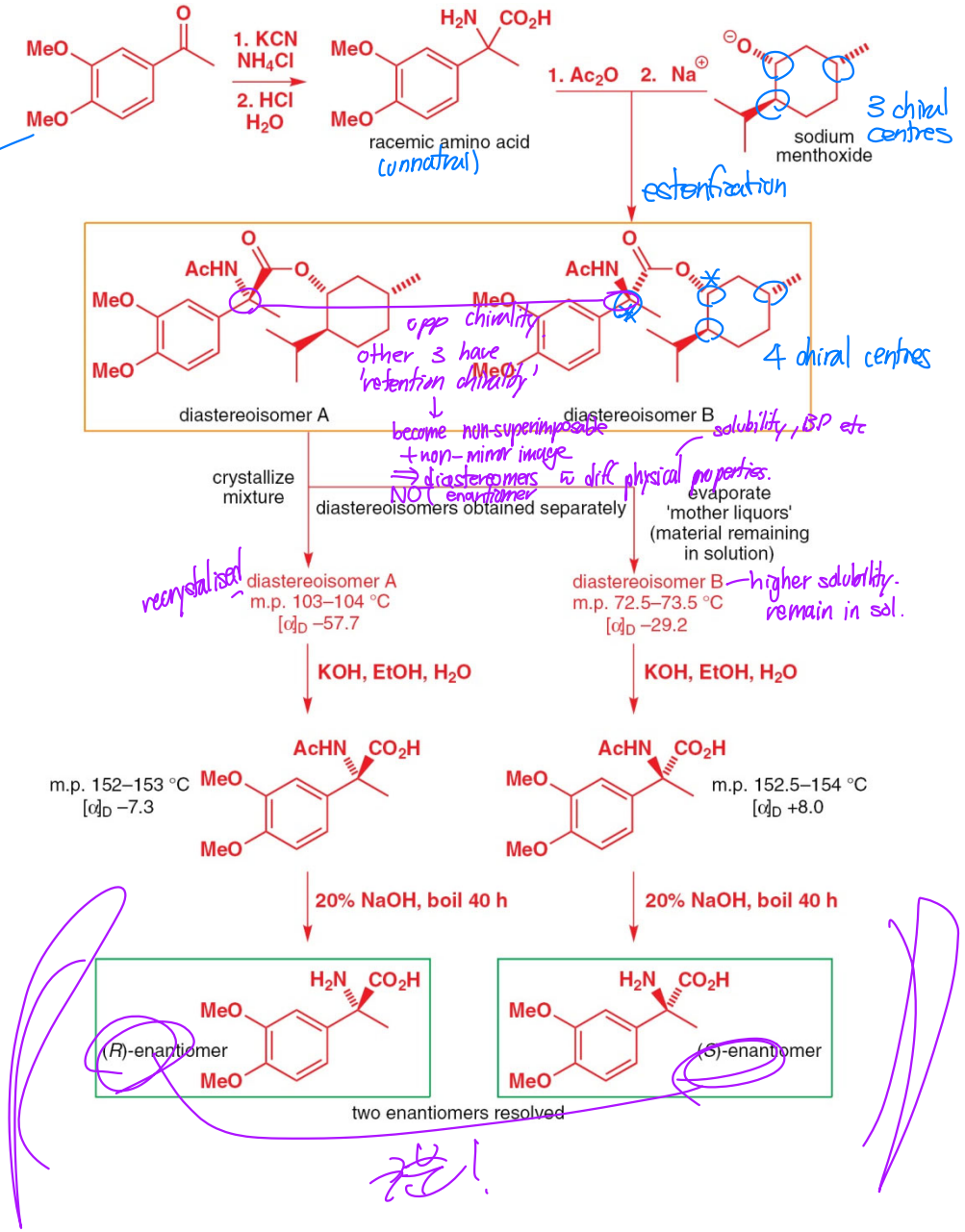


k oxidise alcohol

How to obtain these chiral molecules?

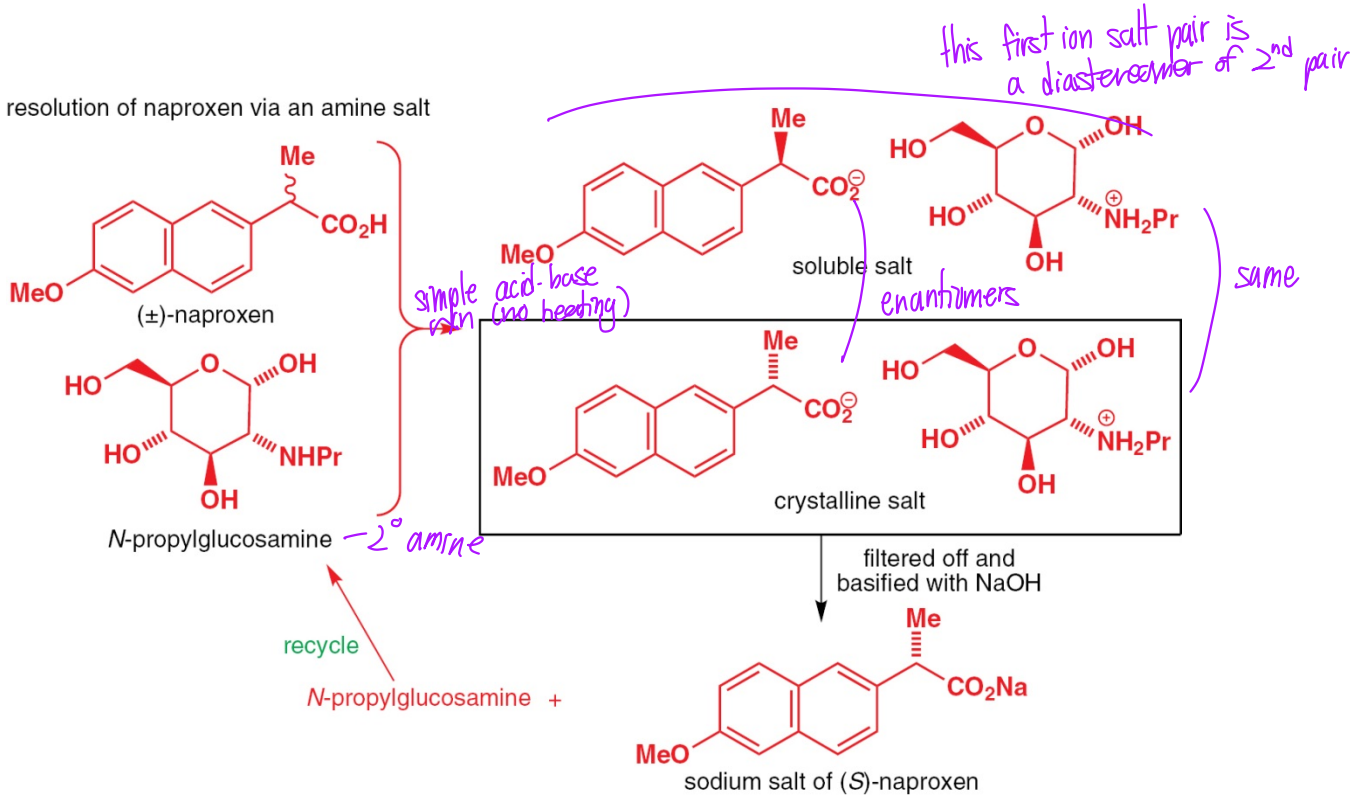
- Resolution





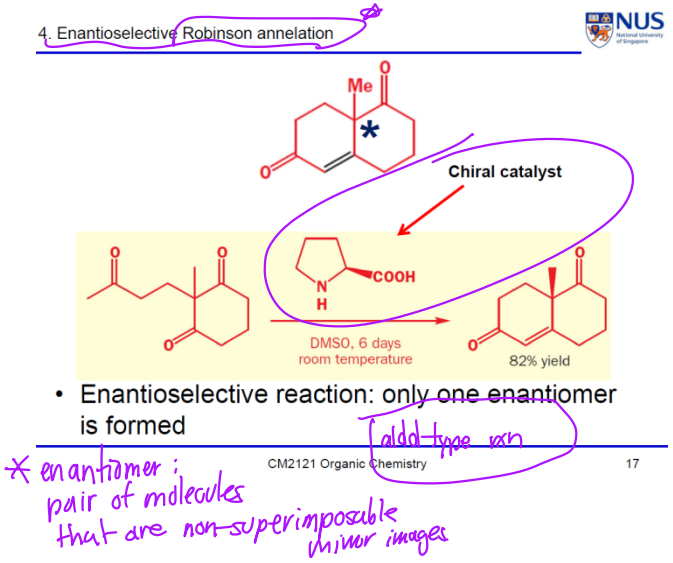
How to obtain these chiral molecules?

- Resolution



Asymmetric catalysis

- Asymmetric reduction of ketones
- Asymmetric hydrogenation of alkenes
- Asymmetric epoxidation → exp 1
- Asymmetric dihydroxylation (Sharpless)
- Asymmetric conjugate addition
- Asymmetric aldol reaction
- …… and many more



Nobel Prize in Asymmetric Catalysis

The Nobel Prize in Chemistry 2021



Ill. Niklas Elmehed © Nobel Prize Outreach.
Benjamin List
Prize share: 1/2



Ill. Niklas Elmehed © Nobel Prize Outreach.
David W.C. MacMillan
Prize share: 1/2

The Nobel Prize in Chemistry 2021 was awarded jointly to Benjamin List and David W.C. MacMillan "for the development of asymmetric organocatalysis."

The Nobel Prize in Chemistry 2001

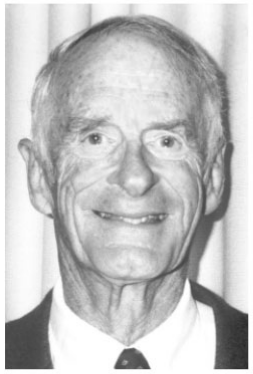


Photo from the Nobel Foundation archive.
William S. Knowles
Prize share: 1/4



Photo from the Nobel Foundation archive.
Ryoji Noyori
Prize share: 1/4

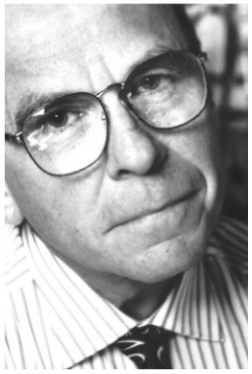
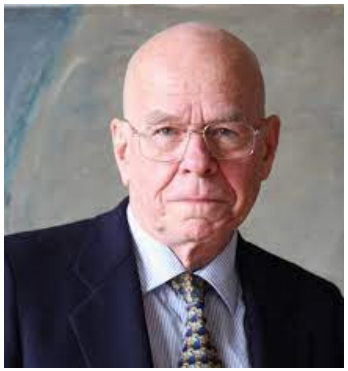
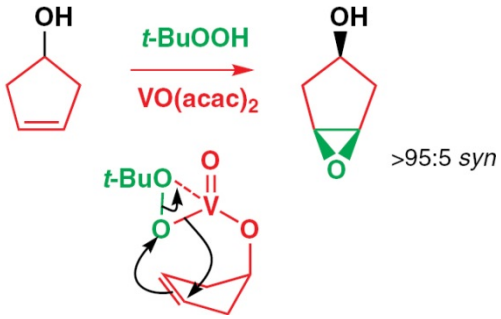


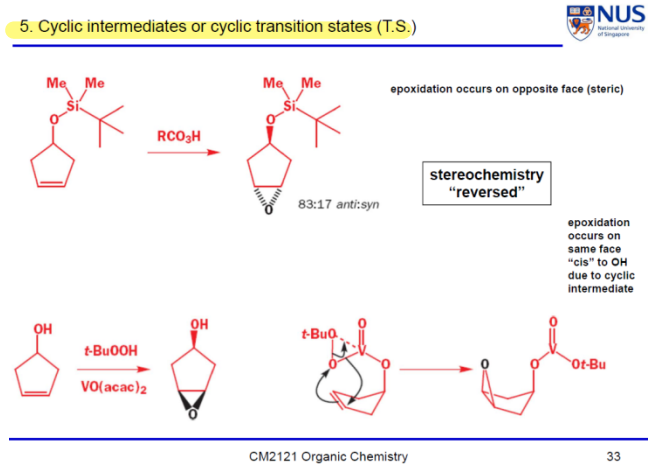
Photo from the Nobel Foundation archive.
K. Barry Sharpless
Prize share: 1/2

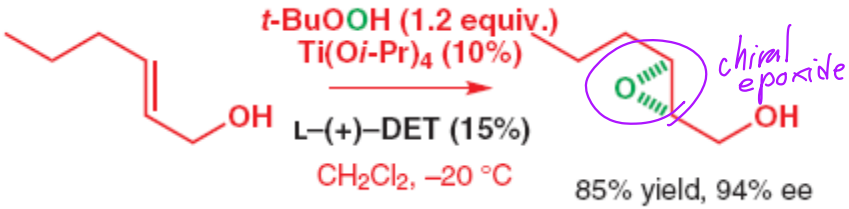
The Nobel Prize in Chemistry 2001 was divided, one half jointly to William S. Knowles and Ryoji Noyori "for their work on chirally catalysed hydrogenation reactions" and the other half to K. Barry Sharpless "for his work on chirally catalysed oxidation reactions."

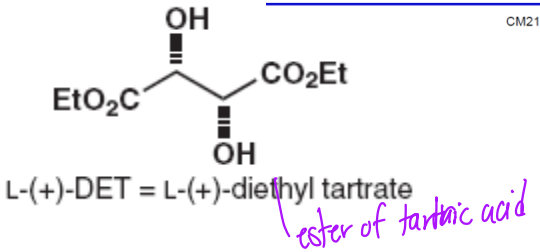
Asymmetric Epoxidation





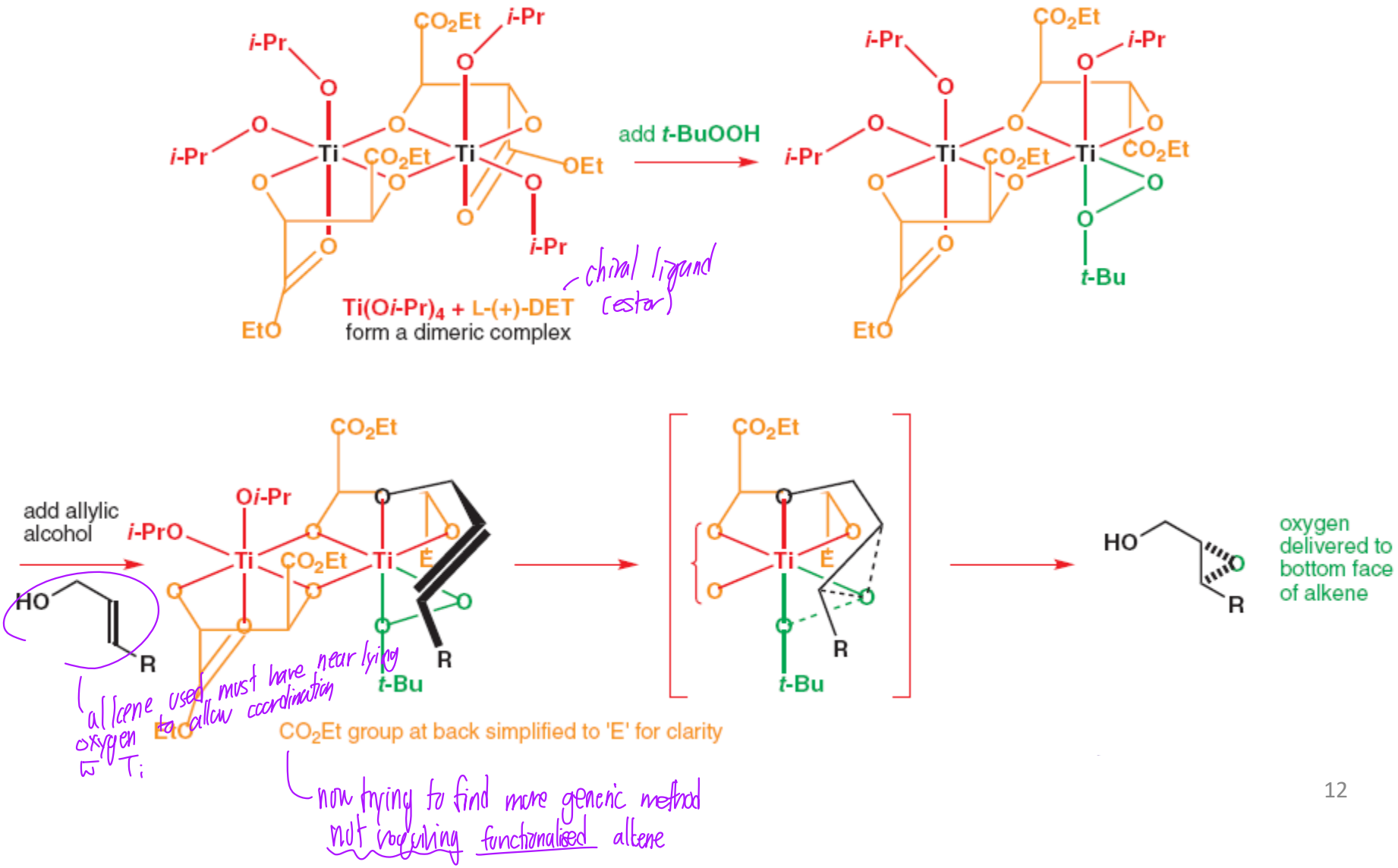






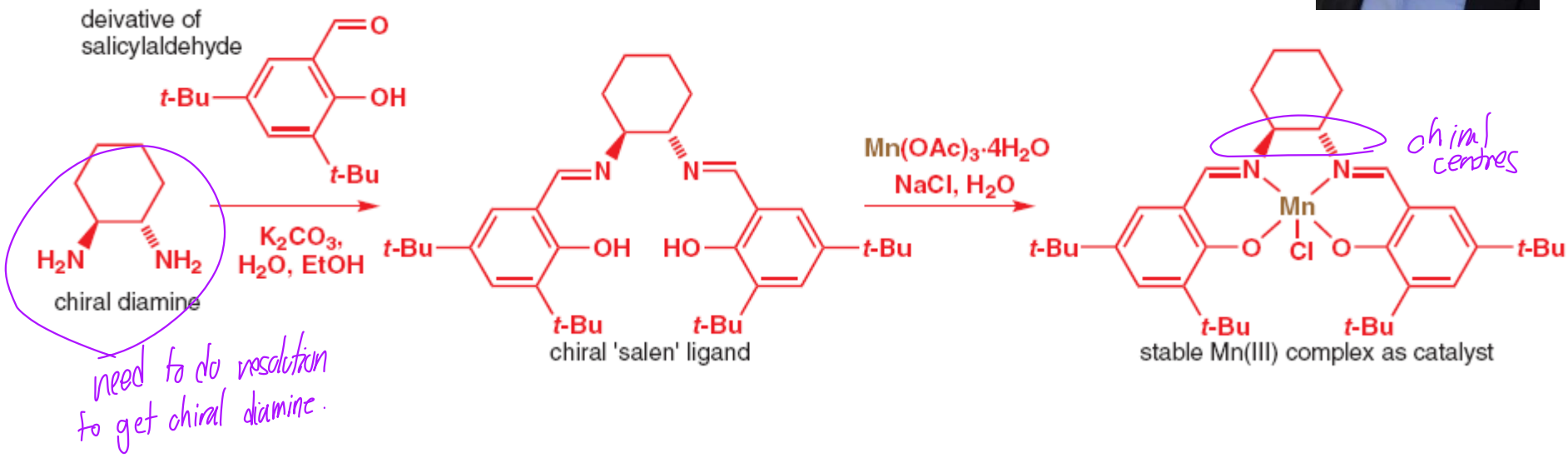
- The first practical method for asymmetric epoxidation, Tsutomu Katsuki and K. Barry Sharpless, *J. Am. Chem. Soc.*, **1980**, *102* (18), 5974–5976

Asymmetric Epoxidation



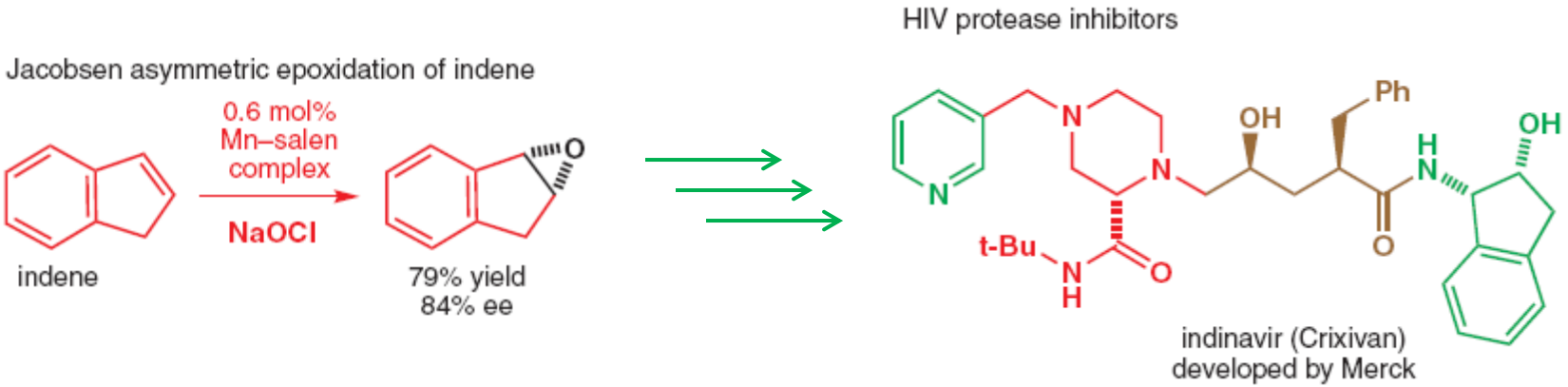
Asymmetric Epoxidation

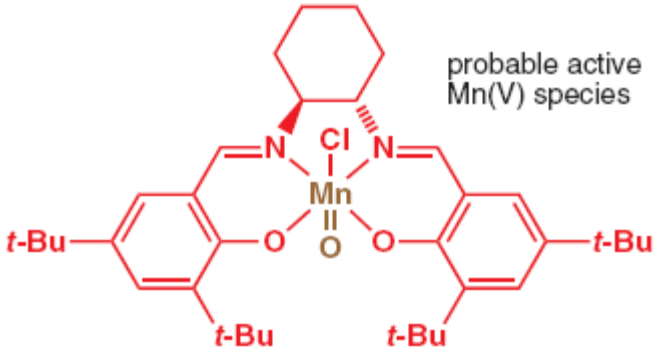




- *Zhang, W.; Loebach, J. L.; Wilson, S. R.; Jacobsen, E. N. J. Am. Chem. Soc.*, **1990**, *112 (7), 2801–2803.*
- *Jacobsen, E. N.; Zhang, W.; Muci, A. R.; Ecker, J. R.; Deng, L. J. Am. Chem. Soc.*, **1991**, *113 (18), 7063-7064.*

Asymmetric Epoxidation





- Jacobsen's catalyst
  - Works on "simple" alkenes without a functional group

Part 2
Practical

Resolution of amine (1)

150.087 g/mol
= 3.752175 g

How do you measure mmol?

Part 1: Resolution of 1,2-diaminocyclohexane

In a 100 mL round bottom flask, L-(+)-tartaric acid (25 mmol) is dissolved in 15 mL of deionized water with stirring. 6.1 mL of 1,2-diaminocyclohexane (a mixture of *cis* and *trans* isomers) is added dropwise. [Note: is the process endothermic or exothermic?] The solution may turn cloudy but complete dissolution is observed within a few minutes. If not, gently heat the solution (if necessary, add a small amount of water) to make the solution clear.

water as solvent
not so vigorous
↓
no need condenser
(보통 heating 자체가 필요 x)

Add condenser when heating

check melting point

Add water gradually, you should observe the solid gradually dissolving → become clear

don't add too much water

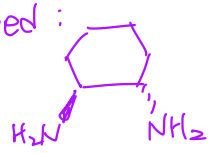
The trans:cis ratio is not provided. Use 50:50 for your calculation, if needed.

molecules actually have > chiral centres.

is there any enantiomer? ←

use solubility diff

need :



→ is this trans or cis?
∴ trans

obs: pale, opaque yellow
→ yellow transparent
(exothermic)

Resolution of amine (2)

Be careful! Glacial acetic acid is corrosive!

After dissolution is complete, 2.5 mL of glacial acetic acid is added in one portion. Observe the formation of precipitation. The reaction mixture is cooled in an ice water bath for at least 30 minutes and product is isolated by suction filtration. The solid is washed with 5.0 mL of cold water and 4 × 5 mL of room temperature methanol.

Do not disturb.

Allow crystals to grow.

If your yield is high, then 5 mL may not be sufficient to fully immerse and wash your product → you can use more solvents to wash.

Also, washing 4 times may not be sufficient to remove remaining starting materials/ impurities → wash more times

Resolution of amine (3)

A common practice for recrystallization. But in this case, slightly excess of water does not affect yield too much.
Amount may be reasonably large, since solvent is water instead of organic solvent.

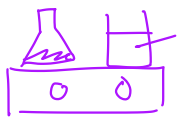
Solubility increases drastically with temperature. Hot water (70-80 °C) may not be enough. Boiling or near boiling (90-95 °C) is needed.
No need to use thermometer. Watch the bubbling.
Add boiling chip before heating.

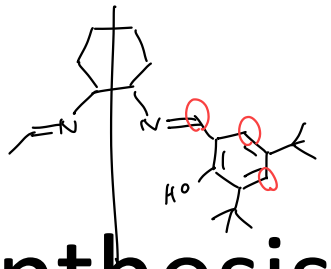
Product is purified by recrystallization from **minimum amount** of boiling water. If there is any insoluble solid or foam, conduct a hot gravity filtration. Allow the solution to cool and collect the crystals by suction filtration. If crystallization does not occur, a small amount (1-2 mL) of methanol can be added to reduce the solubility. A second crop of crystals can usually be collected by adding more methanol to the filtrate.

How to conduct hot gravity filtration?

What is the principle of recrystallization?

Depends on how much water you have used.
You can add more MeOH if needed.





Synthesis of Jacobsen's Ligand (1)

Part 2: Synthesis of Jacobsen ligand [(R,R)-N,N'-bis(3,5-di-tert-butylsalicylidene)-1,2-cyclohexanediamine]

If your product is less, scale down accordingly

264.28 g·mol⁻¹ (0.6753 g → 2.555 mmol) 4.1987 mmol

Towards a 100 mL round bottom flask is added *(R,R)*-1,2-diaminocyclohexane mono-(+)-tartrate salt (2.10 mmol), potassium carbonate (0.58 grams) and water (6.0 mL). The mixture is stirred until dissolution is achieved, and then 22 mL of ethanol is added and reflux is started. In a beaker, 3,5-di-*tert*-butylsalicylaldehyde (4.25 mmol) is dissolved in 10 mL ethanol with careful heating. The aldehyde solution is added to the round bottom flask with a glass dropper. The dropper and beaker are then rinsed with 2.0 mL of hot ethanol. The reaction mixture is refluxed for 1 hour. [Note: the reaction mixture may become quite difficult to stir. Add some solvent if necessary.]

(2 eq · 138.205 g·mol⁻¹, 0.7063 g)

78.37 °C

use condenser at this point

Can add via top opening of condenser

0.9954 g

2.02 eq → 5.171 mmol (234.339 g·mol⁻¹) → 1.2118 g

if × work ① add solvent & 攪拌 to get suspension

② If stirring is still difficult, then can use glass rod to break the large chunk of solid.

⇒ suspension of solid

Synthesis of Jacobsen's Ligand (2)

If your yield is high, 5 mL may not be enough. You can increase the volume if needed.

Water (6.0 mL) is added and the mixture is cooled in ice water bath for 30 minutes. The yellow solid is collected by suction filtration and washed with about 5 mL of ethanol. The solid is dissolved in 25 mL of CH_2Cl_2 and extracted with water (5 mL × 2) and 5 mL of saturated NaCl solution. The organic layer is dried by Na_2SO_4 and the drying agent is removed by filtration. Solvent is removed by rotary evaporator to form the product as a yellow solid.

separating funnel

1) check if valve & cap × leak
   ↳ using DI water

To measure the yield, you need to weigh the empty RBF **before** rotavap

CH_2Cl_2 has low boiling point, so bumping can be very easy and vigorous. Be careful!

Synthesis of Jacobsen's Ligand (2)

Remember to weigh the empty RBF before rotavap!

The product mass is slightly more than the polarimetry amount, so just wash and clean the RBF after polarimetry and NMR.

no need to submit sample

Record the mass of the purified product.

The product (Jacobsen ligand) is characterized by ^{1}H NMR and polarimetry.

To prepare ^{1}H NMR sample:

1. Measure ~ 10 mg of your product into a test tube.
2. Add ~ 0.6 mL $CDCl_3$ solvent and dissolve the product.
3. Transfer the solution into NMR tube and cap properly.
4. Label the NMR tube by the serial number.

Follow the instruction in the lab

Enantiomeric Excess (ee)

- A typical characterization of chiral compounds
- Definition: the excess of one enantiomer over the other, expressed as a percentage of the whole:

$$ee = \frac{|R - S|}{R + S} \times 100\%$$

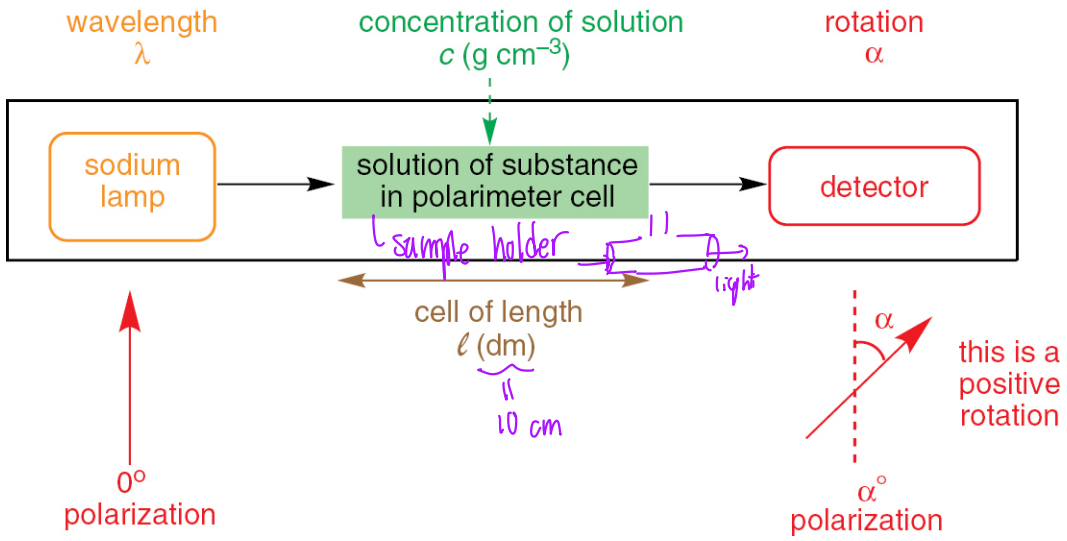
if ee = 100%. only 1 enantiomer. pure.

can use mass, mol or volume as enantiomers have same physical properties.

- Measurement
  - By polarimetry
  - By chromatography: chiral HPLC/GC
  - By spectroscopy

(largely replaced w chiral HPLC

Polarimetry (1)



- Angle of rotation depends on path length, concentration, temperature, solvent and wavelength.

follow literature (DCM) in this case.

Polarimetry (2)

Why accurate? Do you need to know the accurate concentration?

The volume of the cell is about 5-6 mL. So you have about 3 mL solution to rinse the cell before transferring you solution for measurement.

The optical activity of prepared Jacobsen ligand [*(R,R)*-*N,N'*-bis(3,5-di-*tert*-butylsalicylidene)-1,2-cyclohexanediamine] is measured, in order to determine its optical purity. Weigh accurately 0.5-1.0 g of your Jacobsen ligand and then dissolve it in CH_2Cl_2, in a 10 mL volumetric flask. Determine the optical rotation of this solution in a 1 dm cell. Ensure there is no air bubble in the cell and minimize the evaporation of CH_2Cl_2 solvent. After the measurement, discard the CH_2Cl_2 solution (from the cell and the volumetric flask) to designated container and clean up the glassware. Calculate the specific rotation of your sample and compare with the reported value: $[\alpha]_D^{20} = -315°$

You need to use DCM to clean the cell, before passing it to the next user.
Clean the volumetric flask as well.



Polarimetry (3)

- Typical condition:
  - 20 °C
  - Light from sodium lamp at 589 nm – D line of sodium
- Under such "standard" condition, angle of rotation α is proportional to path length *l* (in dm) and concentration c (in g cm^{-3}):

$$\alpha = [\alpha]cl$$

- [α] is called specific rotation and it is independent of c and l. It is the property of the chiral compound.
- Compare $[\alpha]_{experimental}$ with $[\alpha]_{lit}$ → optical purity

References

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